

Aerobic Oxidation of PMB Ethers to Carboxylic Acids

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The first aerobic protocol of direct transformation of *p*-methoxybenzyl (PMB) ethers to carboxylic acids efficiently with $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and TEMPO as catalysts at room temperature has been developed. The reaction accommodates C–Br bond, terminal/non-terminal C–C triple bond, amide, cyano, nitro, ester, and trifluoromethyl groups. Even highly selective oxida-

tive deprotection of different benzylic PMB ethers has been realized. The reaction has been successfully applied to the total synthesis of natural product, (*R*)-6-hydroxy-7,9-octadecadiynoic acid, demonstrating the practicality of the method. Based on experimental studies, a possible mechanism involving oxygen-stabilized benzylic cation has been proposed.

Introduction

Applying protecting groups is a common strategy in multi-step syntheses. *p*-Methoxybenzyl (PMB) ether is one of the most frequently applied protecting strategies for alcohol protection attributing to its high chemical stability in the sequential protection and targeted transformation as well as the easiness of deprotection under mild conditions (Scheme 1a).^[1] On the other hand, direct transformations of PMB ethers without deprotection step to other targeted synthetically versatile functionalities are of high interest for the sake of step-economy. Known reports are on the direct conversions of such PMB ethers to aldehydes in a deprotective manner^[2] and to the synthetically less attractive *p*-methoxy-1-benzoates by the *p*-methoxybenzylic oxygenation^[3] under aerobic conditions (Scheme 1b), however, their direct conversion to carboxylic acids^[4] has never been achieved. Herein, we report a very mild $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO-cocatalyzed direct deprotective aerobic oxidation of PMB ethers to carboxylic acids efficiently (Scheme 1c).

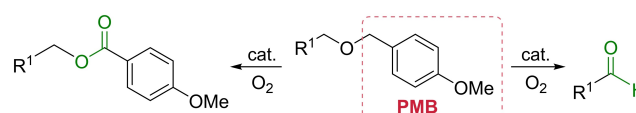
Results and Discussion

We chose the PMB ether of C12 alcohol **1a** as the model substrate. After trial and error, **1a** was successfully converted to the C12 carboxylic acid **2a** in 94% NMR yield and interestingly the PMB part was formed as the aldehyde, 4-methoxybenzaldehyde **4a**, in 93% NMR yield with 10 mol% each of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO, and KCl in DCE at room temperature

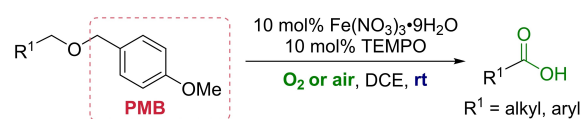
(a) Traditional approach



(b) Catalytic aerobic oxidations of PMB ethers to esters and aldehydes



(c) This work: Iron/TEMPO-cocatalyzed aerobic oxidation of PMB ethers to carboxylic acids



Compatible FG: bromo, terminal/non terminal C–C triple bond, amide, cyano, nitro, trifluoromethyl, and ester group

Scheme 1. The traditional protection strategy for the reaction of alcohols and catalytic aerobic oxidations of PMB ethers.

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($\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO, and KCl in DCE at room temperature (Table 1, entry 1). Using DCM as the solvent gave a similar result (91% NMR yield of **2a**, Table 1, entry 2). The reaction in MeCN afforded a mixture of C12 carboxylic acid **2a** and the corresponding C12 aldehyde **3a**^[2] (Table 1, entry 3). Reducing the loading of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ or TEMPO led to a lower yield of **2a** (Table 1, entries 4 and 5). Control experiment showed that the reaction worked more efficiently *without* KCl within a shortened time of 8 h (Table 1, entry 7). In the absence of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ or TEMPO, no reaction occurred (Table 1, entries 8 and 9). The reaction with 4-OH-TEMPO replacing TEMPO is much slower (Table 1, entry 10).

Subsequently, we examined the scope of the $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO-cocatalyzed aerobic oxidative process. As shown in Scheme 2, PMB ethers with various length of alkyl chains for the R group gave the corresponding carboxylic acids **2a–c** in generally decent yields. Two mmol scale reaction of **1a** could successfully afford the corresponding acid **2a** in a high yield of 95%. The reaction could also be performed by using an

Table 1. Optimization of the reaction conditions.^[a]

Entry	X/Y/Z	Solvent	t (h)	Recovery of 1a (%)	NMR yield (%)		
					2a	3a	4a ^[b]
1	10/10/10	DCE	18	0	94	0	93 (5)
2	10/10/10	DCM	18	0	91	0	90 (2)
3	10/10/10	MeCN	18	0	29	56	99
4	5/10/10	DCE	18	0	80	9	99
5	10/5/10	DCE	18	0	90	5	99
6	10/10/5	DCE	12	0	97	0	93 (5)
7	10/10/0	DCE	8	0	95	0	93 (3)
8	10/0/0	DCE	8	100	0	0	0
9	0/10/0	DCE	8	100	0	0	0
10 ^[c]	10/10/0	DCE	8	17	6	62	82

[a] Reaction conditions: **1a** (0.5 mmol), Fe(NO₃)₃·9H₂O (X mol%), TEMPO (Y mol%), and KCl (Z mol%) in solvent (1.0 mL) with an oxygen balloon at rt. The recovery and NMR yield were determined by ¹H NMR analysis of the crude product. [b] NMR yield of 4-methoxybenzoic acid **5a** is shown in the parenthesis. [c] 4-OH-TEMPO was used instead of TEMPO.

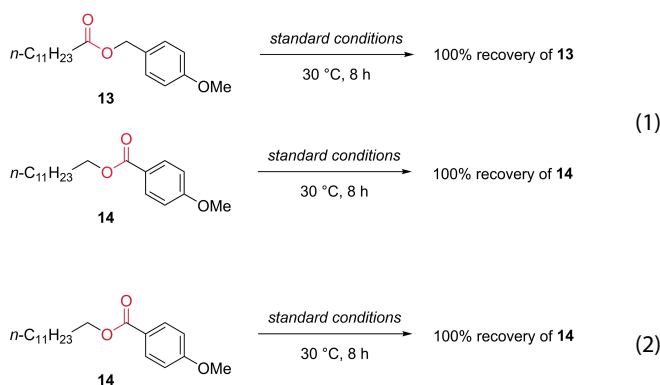
air balloon for 23 h with a yield of 89%. The substrate with a highly sterically hindering group such as adamantyl also worked (**2d**). Synthetically versatile functional groups, such as bromo, terminal/non-terminal C–C triple bond, and phenyl group, were also applicable (**2e–h**). Importantly, the substrates derived from pharmaceuticals and natural products including felbinac (**2k**), naproxen (**2l**), ibuprofen (**2m**) and sclareolide (**2n**), also proceeded well. Furthermore, the optically active substrates worked without any racemization [(*S*)-**2h**, (*S*)- or (*R*)-**2i**, (*S*)- or (*R*)-**2j**, (*S*)-**2l**, (*S*)-**2m**]. Optically active tricyclic lactone sclareolide **2n** was also formed in 80% yield from PMB ether **1n** bearing a free tertiary hydroxy group. Last but not least, mono- and bis- PMB ethers of 1,8-octanediol, **1p** and **1o**, both afforded diacid **2o**.

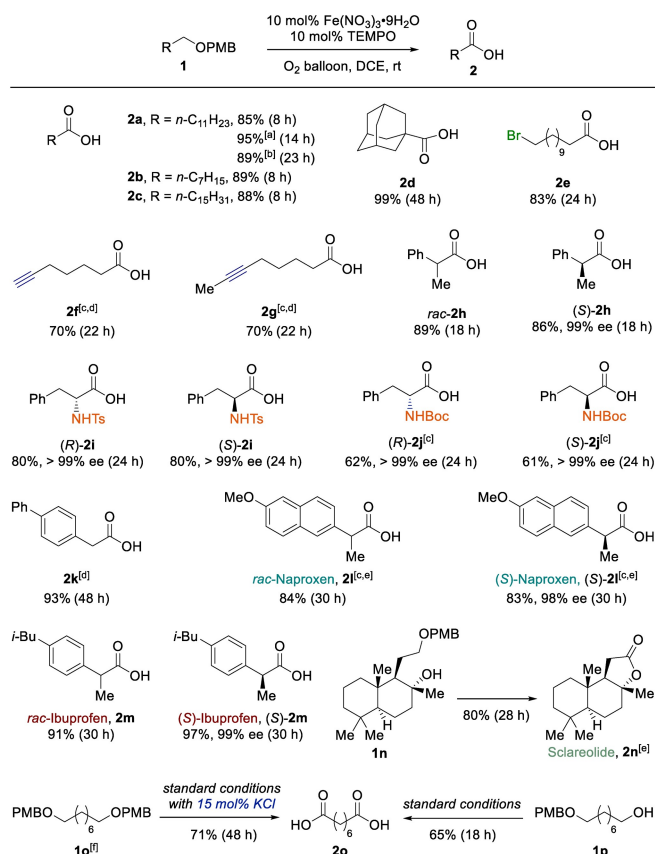
We further scrutinized the substrate scope of the rather challenging differently substituted benzylic PMB ethers. As shown in Scheme 3, the electron-withdrawing and potentially useful groups such as cyano, nitro, trifluoromethyl, or ester groups are all compatible affording the non-methoxy-substituted benzoic acids **2r–u** highly selectively in good yields using 15 mol% each of Fe(NO₃)₃·9H₂O and TEMPO. When there is a *meta*-nitro group, extending the reaction time to 72 h was required affording the *m*-nitrobenzoic acid **2v**. Notably, by using 20 mol% each of Fe(NO₃)₃·9H₂O and TEMPO, benzylic PMB ether **1q** itself could also react smoothly to give benzoic acid **2q** in 79% yield within 48 h, accompanied by just 6% benzaldehyde **3q**. Moreover, under standard conditions, for 1,6-diol substrates with one hydroxy protected as PMB ether and the other protected as *p*-CN (**1w**) or *p*-CF₃ (**1x**) benzylic ether, the PMB ether was highly selectively removed affording **2w** and **2x** with the *p*-CN (**1w**) or *p*-CF₃ (**1x**) benzylic ether untouched due to the much lower reactivity of electron-

withdrawing benzylic ethers under the current aerobic conditions.

In order to demonstrate the practicality of this method, the natural product (*R*)-**12**^[5] was synthesized in 8 steps from the commercially available 1,6-hexanediol (Scheme 4): Aldehyde **8** was obtained from 1,6-hexanediol **6** via protection and oxidation in 74% and 76% yields, respectively. By applying sequential alkynylation, two consecutive Novozym 435-catalyzed kinetic resolutions,^[6] and acetylation, propargylic acetate (*R*)-**10** was obtained in > 99% ee. The current aerobic oxidation of (*R*)-**10** successfully afforded (*R*)-**11** in 97% NMR yield under the standard conditions, which was submitted to next step without further purification. Finally, the copper-catalyzed coupling of (*R*)-**11** with alkynyl bromide yielded (*R*)-**12** in 88% yield with 95% ee.^[7]

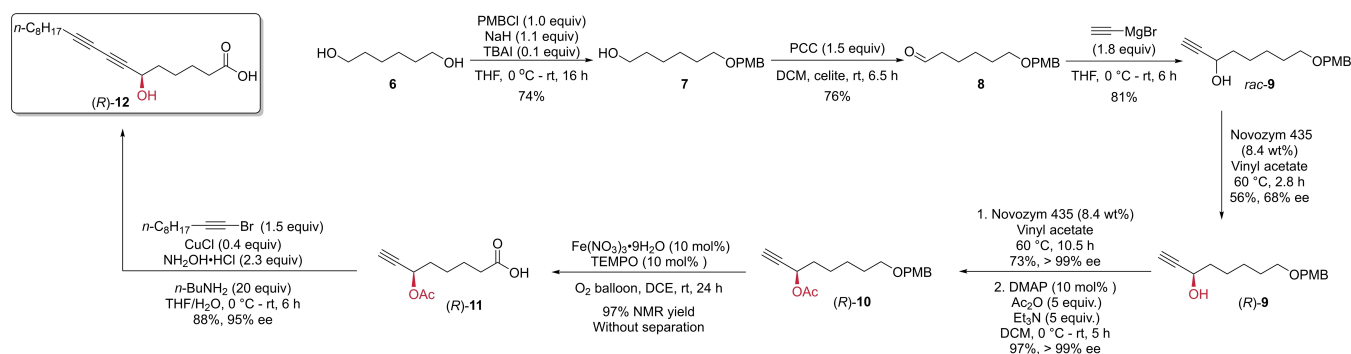
To gain insight into the mechanism, the corresponding esters **13**^[8] and **14**^[9] were synthesized deliberately and submitted to the standard conditions. Interestingly no reaction was observed, excluding the intermediacy of these two esters (Equations (1) and (2)).



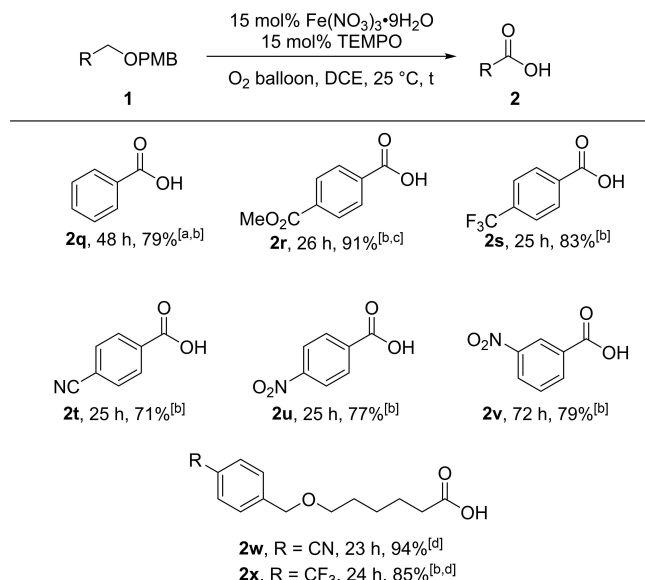


Scheme 2. Scope study. Reaction conditions: **1** (0.5 mmol), Fe(NO₃)₃·9H₂O (10 mol%), and TEMPO (10 mol%) in DCE (1.0 mL) with an oxygen balloon at rt. All yields were isolated yields. [a] 2 mmol of **1a** were used. [b] Air balloon was used instead of oxygen balloon and the reaction was conducted at 25 °C. [c] The acid was not easy to isolate. We characterized the corresponding methyl ester after esterification with TMSCHN₂. [d] 10 mol% of KCl was added. [e] 2.0 mL of DCE were used. [f] **1o** (1.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), TEMPO (10 mol%), and KCl (15 mol%) in DCE (4.0 mL) with an oxygen balloon at rt.

Based on these results and the literature,^[10–12] a possible mechanism is proposed as shown in Scheme 5: •NO₂ and •NO have been reported to be readily generated via the interaction of Fe(NO₃)₃·9H₂O with TEMPO.^[13] •NO₂ would abstract the benzylic hydrogen atom producing benzylic radical **Int 1**.^[10]



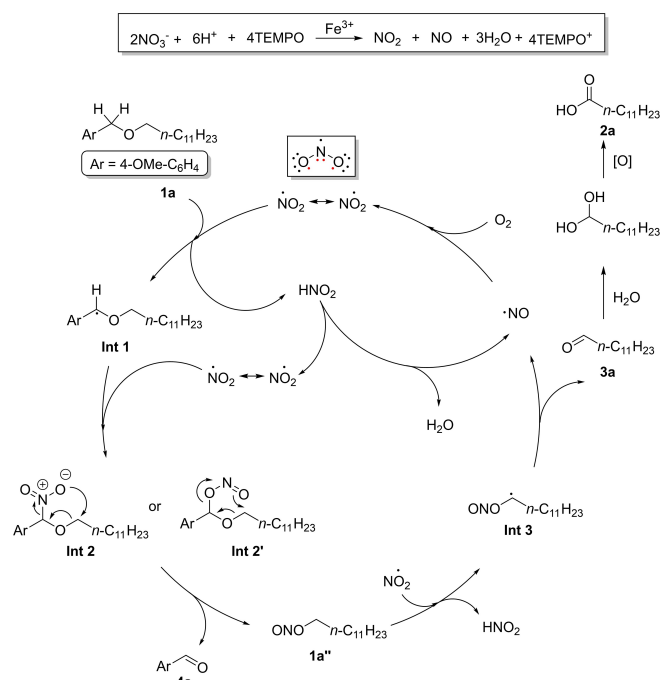
Scheme 4. Total synthesis of (R)-6-hydroxy-7,9-octadecadiynoic acid (**R**)-**12**.



Scheme 3. Scope of aerobic oxidation of benzylic PMB ethers and selective aerobic oxidation of different benzylic ethers. Reaction conditions: **1** (0.5 mmol), Fe(NO₃)₃·9H₂O (15 mol%), and TEMPO (15 mol%) in DCE (1.0 mL) with an oxygen balloon at 25 °C. All yields were isolated yields unless otherwise mentioned. [a] Fe(NO₃)₃·9H₂O (20 mol%) and TEMPO (20 mol%) were used; 6% NMR yield of benzaldehyde **3q** was determined by ¹H NMR analysis of the crude product. [b] The acid was not easy to isolate. We characterized the corresponding methyl ester after esterification with TMSCHN₂. [c] 2% NMR yield of 4-(methoxycarbonyl)benzaldehyde **3r** was determined by ¹H NMR analysis of the crude product. [d] Fe(NO₃)₃·9H₂O (10 mol%) and TEMPO (10 mol%) were used.

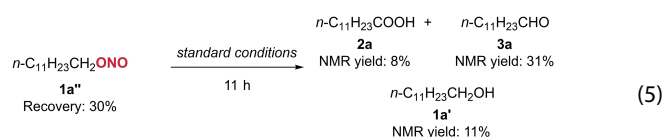
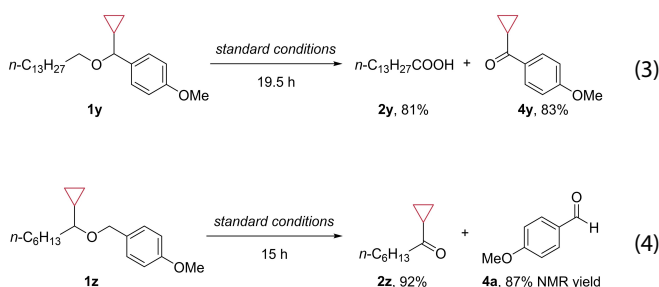
Next, **Int 1** may couple with •NO₂ affording nitro ester **Int 2** or **Int 2'**, which would undergo rearrangement to afford 4-methoxybenzaldehyde **4a** and nitrite **1a''**.^[10,11] With the help of •NO₂, the transformation of nitrite **1a''** to nitrite radical **Int 3** leads to aldehyde **3a** by the elimination of •NO radical from the **Int 3**.^[12] Upon further oxidation, carboxylic acid product **2a** would be delivered.

In order to see whether mechanism A is rational, substrates **1y** and **1z** each with a cyclopropyl unit were synthesized and submitted to the standard conditions. No ring opening products were observed (Equations (3) and (4)). Moreover, nitrite **1a''** was recovered in 30% yield with only 8% yield of **2a**



Scheme 5. The possible mechanism A.

and 31% of aldehyde **4a** under the standard conditions (Equation (5)), excluding the intermediacy of both benzylic radical **Int 1** and the nitrite **1a''**. These results rule out mechanism A.



Subsequently, $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO-catalyzed and NO_2 /TEMPO-catalyzed aerobic oxidation of **1a** were monitored (Figure 1). We found that 15 mol% of NO_2 and 10 mol% of TEMPO-catalyzed aerobic oxidation of PMB ether matched the data of the current standard conditions of 10 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 10 mol% of TEMPO catalytic recipe (for more details, see SI). Surprisingly, only 6% of 4-methoxybenzaldehyde **4a** was oxidized to 4-methoxybenzoic acid **5a** as observed during the monitoring of the whole reaction process. These results illustrated $\bullet\text{NO}_2$ radical may be involved in this reaction.

Next, the corresponding alcohol **1a'** was submitted to the standard conditions.^[4n] Surprisingly, the reaction was very slow yielding only 55% NMR yield of acid **2a** with 40% NMR yield of aldehyde **3a** even after 11.5 hours. Given that 4-methoxybenzaldehyde **4a** was detected as the product in the current transformation, we reasoned that aldehyde **4a** may accelerate the process of oxidation of alcohol. Indeed, when using 4-methoxybenzaldehyde **4a** (40 mol%) as additive under the standard conditions, the alcohol **1a'** was transformed to acid within 11.5 hours (Equation (6)). We then monitored the reactions of PMB ether and alcohol (Figure 2a and b) and found that indeed the reaction rate of PMB ether **1a** is much faster than that of alcohol **1a'** under the standard conditions. This phenomenon has been further confirmed by monitoring of alcohol **1a'** in the presence of 4-methoxybenzaldehyde **4a** (Figure 2c). In addition, it is also surprising that only 5% of *p*-methoxybenzaldehyde **4a** was oxidized to the corresponding 4-methoxybenzoic acid **5a** (Figure 2c).

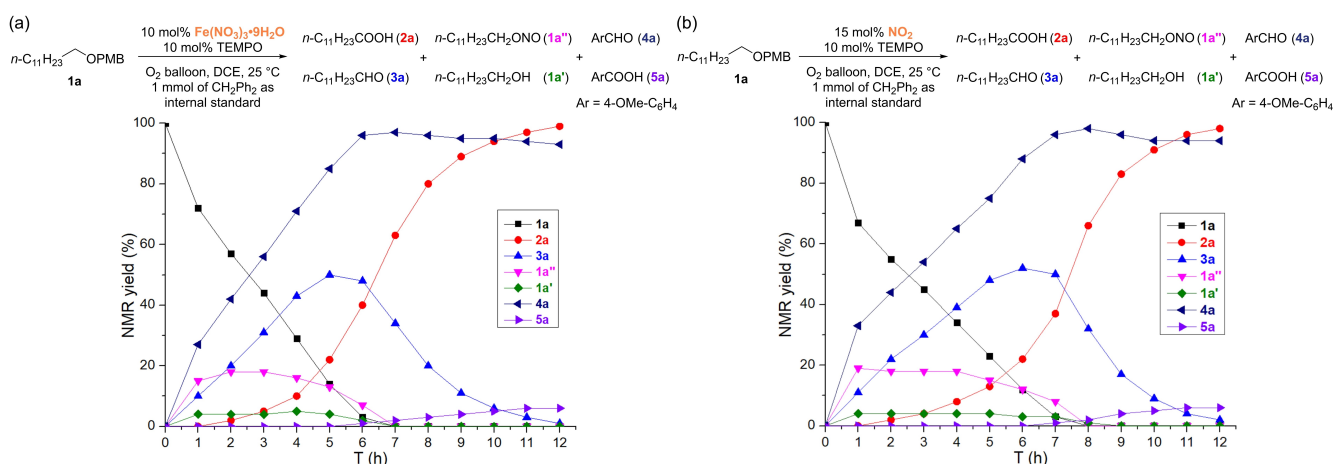
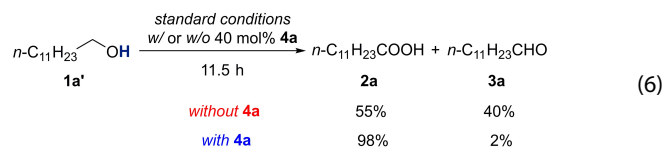


Figure 1. Comparison of catalytic aerobic oxidations of PMB ether **1a**: 10 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ vs. 15 mol% of NO_2 ; (a) 10 mol% of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 10 mol% of TEMPO-catalyzed aerobic oxidation of **1a**; (b) 15 mol% of NO_2 and 10 mol% of TEMPO-catalyzed aerobic oxidation of **1a**.

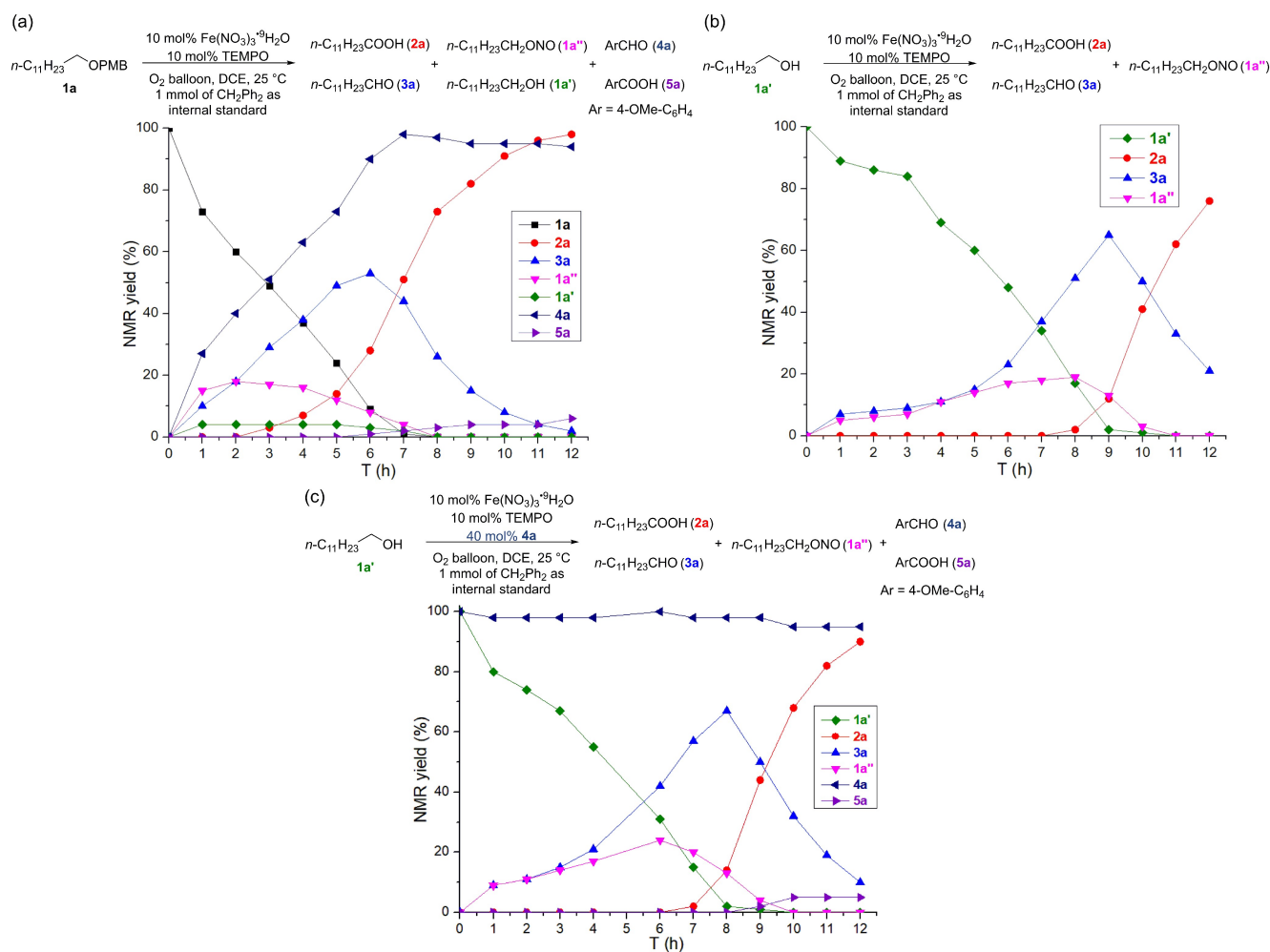
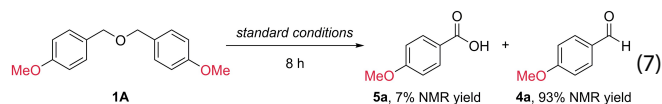


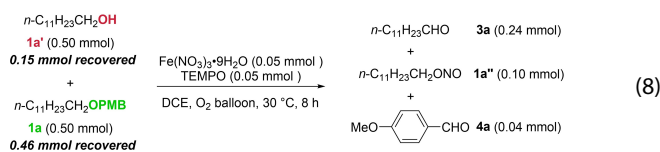
Figure 2. Comparison of the reaction of PMB ether **1a** with that of alcohol **1a'**; (a) 10 mol% of Fe(NO₃)₃·9H₂O and 10 mol% of TEMPO-catalyzed aerobic oxidation of **1a**; (b) 10 mol% of Fe(NO₃)₃·9H₂O and 10 mol% of TEMPO-catalyzed aerobic oxidation of **1a'**; (c) 10 mol% of Fe(NO₃)₃·9H₂O, 10 mol% of TEMPO, and 40 mol% of **4a**-catalyzed aerobic oxidation of **1a'**.

This was further confirmed by the fact that under the standard conditions, the reaction of *p*-methoxybenzyl PMB ether **1A** afforded 4-methoxybenzoic acid **5a** only 7% NMR yield with 93% NMR yield of 4-methoxybenzaldehyde **4a**, which is matched with the results in monitoring that only 6% of 4-methoxybenzoic acid **5a** was detected (Equation (7) vs. Figure 1).



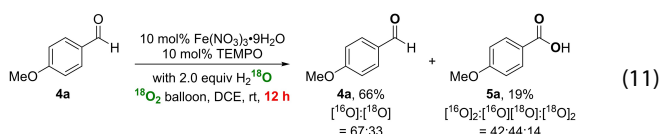
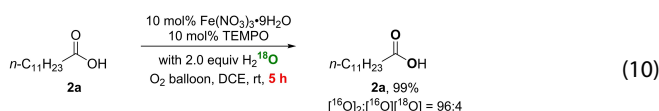
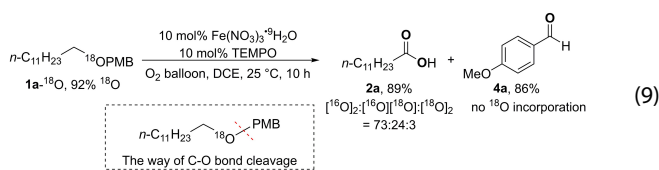
We also set up the competitive experiment of alcohol **1a'** (0.50 mmol) and PMB ether **1a** (0.50 mmol) with 0.05 mmol each of Fe(NO₃)₃·9H₂O and TEMPO as catalysts (Equation (8)): 0.24 mmol of aldehyde **3a**, 0.10 mmol of nitrite **1a''**, and 0.04 mmol of 4-methoxybenzaldehyde **4a** were afforded after 8 h. The amounts of recovered alcohol **1a'** and PMB ether **1a** were 0.15 mmol and 0.46 mmol, respectively. These results demonstrated that the alcohol **1a'** was much more reactive

when the alcohol and the PMB ether both existed, which was consistent with the observation in the monitoring experiment (Figure 1a) that the concentration of alcohol was low during the reaction.

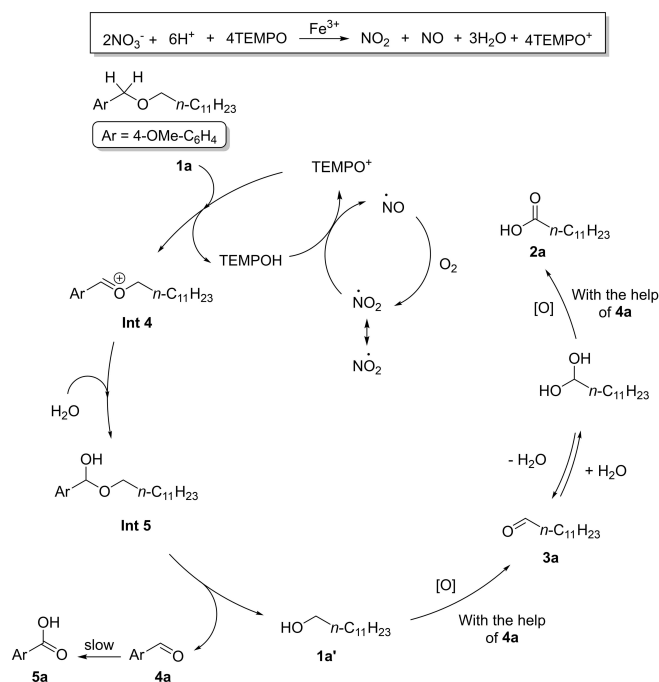


In addition, isotopic ¹⁸O labeled substrate **1a**-¹⁸O was submitted to the standard conditions, the content of isotopic ¹⁸O in the acid **2a** is much higher than 4-methoxybenzaldehyde **4a** (Equation (9)). Oxygen atom exchange experiments of **2a** or **4a** with H₂¹⁸O were also conducted: Oxygen in the acid **2a** remained almost unchanged (Equation (10)), however, the oxygen in 4-methoxybenzaldehyde **4a** exchanged with H₂¹⁸O affording 33% ¹⁸O labeled 4-methoxybenzaldehyde under the standard conditions (Equation (11)). Thus, it is concluded that the isotopic ¹⁸O in the acid **2a** came from **1a**-¹⁸O (Equation (9)).

All of these results indicated the cleavage of C–O bond connected to PMB group (Equation (9)).



Based on these data, an alternative mechanism is proposed (Scheme 6): the oxoammonium cation $\text{TEMPO}^{+[\text{14}]}$ was generated via the interaction of TEMPO with nitrates in the presence of H^{+} as the catalytically active species. An oxonium ion **Int 4** is formed by hydride transfer from the benzylic position of **1a** to the oxygen atom of $\text{TEMPO}^{+[\text{15}]}$. The in situ generated TEMPOH was subsequently oxidized by $\bullet\text{NO}_2$ to regenerate $\text{TEMPO}^{+[\text{16}]}$. Meanwhile $\bullet\text{NO}_2$ was reduced to $\bullet\text{NO}$, which would be oxidized by oxygen to regenerate $\bullet\text{NO}_2$. **Int 4** would react with water to form the hemiacetal **Int 5**, which would form 4-methoxybenzaldehyde **4a** and alcohol **1a'** upon hydrolysis. The resulting alcohol **1a'** would be further oxidized to give the carboxylic



Scheme 6. The possible mechanism B.

acid product **2a**.^[4n] However, the oxidation of 4-methoxybenzaldehyde **4a** to 4-methoxybenzoic acid **5a** was slow. The role of 4-methoxybenzaldehyde **4a** on the oxidation of alcohol to carboxylic acid is still unclear.

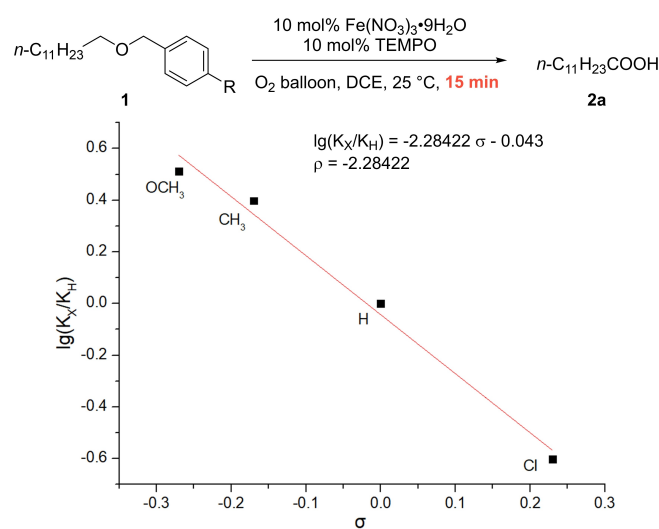
In order to further verify the intermediacy of oxonium ion **Int 4**, we investigated the effect of *para*-substituent group in the phenyl ring of **1** (Table 2). The reactivity of **1** dropped along with the electron-withdrawing ability of this R group: 4-methoxybenzyl ether **1a** was fully consumed in only 8 h (Table 2, entry 1) while the 4-cyanobenzyl substrate **1a-C** was totally inert under the standard conditions (Table 2, entry 4). The results fitted the Hammett plot^[17] of $\log(k_x/k_h)$ versus σ , as shown in Scheme 7. The Hammett slope of $\rho = -2.28422$ suggests the formation of this positively charge species,^[15] further supporting the formation of **Int 4**.

Table 2. The effect of substituent in the benzyl group.^[a]

Reaction scheme showing the oxidation of **1a** (0.5 mmol) to **2a** (85% yield), **4a** (95% NMR yield), and **5a** (2% NMR yield) using 10 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 10 mol% TEMPO in DCE with an oxygen balloon at rt for 8 h.

Entry	R	Recovery of 1 a (%)	NMR yield (%)		
			2 a	3 a	4^[b]
1	OMe (1 a)	0	95	0	93 (3)
2	Me (1 a-A)	11	73	16	88
3	H (1 a-B)	39	45	9	55
4	CN (1 a-C)	97	0	0	0

[a] Reaction conditions: **1** (0.5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (10 mol%), and TEMPO (10 mol%) in DCE (1.0 mL) with an oxygen balloon at rt for 8 h. The recovery and NMR yield were determined by ^1H NMR analysis of the crude product. [b] NMR yield of corresponding benzoic acid of **5** is shown in the parenthesis.

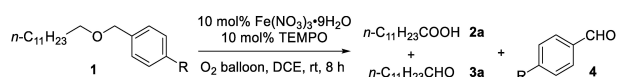


Scheme 7. Hammett plot for the aerobic oxidation of 4-R-substituted benzylic ethers under the standard reaction conditions.

Conclusions

In summary, we have developed an efficient and highly selective aerobic transformation of PMB ethers affording the corresponding carboxylic acids under the catalysis of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO with pure oxygen or air as the green oxidant. The reaction proceeds under very mild reaction conditions and allows transformation of a variety of PMB ethers accommodating many functionalities. No racemization occurred for PMB ethers of optically active alcohols. Moreover, total synthesis of natural product (*R*)-6-hydroxy-7,9-octadecadiynoic acid (*R*)-12 has been realized with this reaction as the key step. Experimental studies indicated a unique mechanism involving oxygen-stabilized benzylic cation. Further studies are being actively pursued in our laboratory.

Experimental Section



$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.9 mg, 0.05 mmol), TEMPO (8.2 mg, 0.05 mmol), and **1a** (153.8 mg, 0.5 mmol) were added to 25 mL flask sequentially. The flask was degassed to remove the air inside completely and refilled with O_2 by an O_2 balloon for three times. After adding DCE (1.0 mL), the resulting mixture was stirred at room temperature for 8 h until completion of the reaction as monitored by TLC (petroleum ether/ethyl acetate=5/1). The resulting mixture was diluted with DCM and filtrated through a short column of silica gel eluted with DCM (120 mL). After evaporation, 95 % NMR yield of **4a** and 2 % NMR yield of **5a** were determined by ^1H NMR analysis of the crude product with CH_2Br_2 (35.0 μL) as the internal standard. After evaporation, an aqueous solution of Na_2CO_3 (pH = 11–12, 10 mL) was added to the crude product **2a**. The aqueous layer was extracted with ethyl acetate (20 mL $\times$ 3). Then an aqueous solution of HCl (3.0 M) was added to the aqueous layer for adjusting pH of the solution to 1–2. The aqueous layer was diluted with ethyl acetate and extracted with ethyl acetate (20 mL $\times$ 3). The organic layer was separated, washed with brine, dried over anhydrous Na_2SO_4 , filtrated, concentrated, and evaporated to remove the solvent to afford **2a**^[4n] (85.0 mg, 85 %) as solid: m.p. 43.1–43.9 °C (petroleum ether/ethyl acetate) (reported:^[4n] m.p. 43–44 °C (petroleum ether/ethyl acetate)); ^1H NMR (400 MHz, CDCl_3): δ 11.87 (brs, 1 H, COOH), 2.34 (t, J =7.4 Hz, 2 H, CH_2), 1.63 (quint, J =7.3 Hz, 2 H, CH_2), 1.42–1.17 (m, 16 H, $8\times\text{CH}_2$), 0.88 (t, J =6.8 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, CDCl_3): δ 180.7, 34.1, 31.9, 29.6, 29.4, 29.3, 29.2, 29.1, 24.6, 22.7, 14.0.

Supporting Information

The authors have cited additional references within the Supporting Information.^[18–57]

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Mr En Zheng in this group for reproducing the results of (*R*)-**2j'**, (*S*)-**2l'** and **2t'** presented in Scheme 2 and Scheme 3.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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